AI Adoption, Labor Market Polarization and Income Inequality: an Agent-Based Model with Imitation Dynamics

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1. Introduction

In the present work, we develop an agent-based model (ABM) to model the process of Artificial Intelligence (AI) adoption by firms and analyze its distributional consequences. The model is calibrated to recent large-scale evidence on early-career employment losses in AI-exposed occupations (Brynjolfsson et al., 2025a). Building on a bipartite firm-worker economy, we extend standard Poisson adoption dynamics to incorporate inter-firm social pressure through an Axelrod network structure, allowing AI adoption to cascade through firm-level peer connections rather than arrive independently. We use the model to trace the resulting trajectories of wealth inequality (measured through the Gini coefficient) and labor market polarization between Junior and Senior workers.

Specifically, this work focuses on modelling the seniority-biased displacement effect and the vulnerability of Junior workers that has been highlighted by recent economic literature on automation and the labour market effects of Generative AI. While in the past the adoption of new technologies, such as ICT revolution and industrial robotics, primarily displaced medium-skilled, routine manual tasks (Autor et al., 2003), recent empirical evidence suggests that AI targets non-routine, cognitive, and desk-based activities (Webb, 2019). Within this kind of tasks, according to recent evidence, early-career profiles appear disproportionately exposed. As conjectured in Brynjolfsson et al. (2025a), indeed, the tasks typically assigned to Junior workers rely heavily on codifiable, entry-level knowledge (basic programming, initial drafting, routine data synthesis, etc.) which falls within the technical capabilities of Generative AI

models. Conversely, Senior workers seem to be protected from immediate displacement. This protection is not driven by formal educational credentials alone, but by the accumulation of firm-specific human capital and tacit knowledge which AI cannot easily replicate or automate. Consequently, in the present work we model AI adoption as an asymmetric labor market shock which disproportionately displaces entry-level workers. Through the model, we aim at capturing how the truncation of early-career employment opportunities may disrupt wealth accumulation paths, feeding directly into structural income polarization and generating a localized poverty trap for the younger generation of workers.

As mentioned, the model is presented in Section 2 and calibrated in Section 3 following the stylised facts found by Brynjolfsson et al. (2025a). Finally, in Section 4 we study the properties of the model in order to answer the following research questions:

1. How does the firm-level staggered adoption of Generative AI drive endogenous polarization and inequality dynamics between Junior and Senior workers?
2. How does the interaction of labor market frictions, structural credit constraints, and the specific timing of displacement following the AI technological shock determine unemployment dynamics, workers' access to retraining, and the allocation of vulnerable workers into a path-dependent poverty trap?

The model is built as an extension of the Boltzmann wealth model, a simple ABM that borrows tools from statistical physics to describe the probability distributions of money, income and wealth in a stylised economy (Dragulescu and Yakovenko, 2002). The Boltzmann approach involves the measurement of the evolution of wealth in a system of initially identical agents that interact through binary monetary transactions (Pareschi and Toscani, 2014). In the standard Boltzmann framework, indeed inequality emerges endogenously from bilateral exchange amongst agents who begin with identical endowments: the random redistribution of wealth generates a stationary distribution that resembles the Boltzmann-Gibbs distribution of energy in statistical physics.

Our model departs from the original Boltzmann framework in two fundamental respects. First, in the baseline model of Section 2.1 we suppress the background stochastic dynamics: the baseline is constructed so that, at the beginning, equal initial endowments and uniform wages constitute a stable fixed point of the system, with no random redistribution between workers. Inequality in our model does not emerge from bilateral exchange but is introduced exclusively by the AI adoption shock and the asymmetric labour market mechanisms it triggers, lately presented in Section 2.2. Second, we replace the homogeneous agent population of the Boltzmann model with a bipartite economy of firms and workers connected through an explicit labour network, allowing us to trace the distributional consequences of a technological shock across demographic groups rather than across an undifferentiated wealth distribution. This controlled departure from the Boltzmann baseline is by design: it ensures that any divergence in the Gini coefficient or the Polarization Index observed in subsequent extensions is attributable exclusively to the mechanisms introduced by AI adoption, and not to background stochastic dynamics inherited from the Boltzmann framework.

2. The Model

2.1. Baseline Model

In what follows we present the agent-based architecture of the model, built in the spirit of the Boltzmann wealth model. In the present Section we introduce the baseline model, which serves as the control scenario of the simulation.

Consider a stylised closed economy populated by initially homogeneous firms and workers which behave according to heuristic rules. In the baseline specification, firms produce output and distribute wages to a fully employed labour force, with no technological shocks and no unemployment. By design, this environment generates no systematic divergence between demographic groups (i.e., Junior *vis-à-vis* Senior workers), as workers begin with identical endowments and receive identical wages. The expected outcomes are therefore a Gini coefficient of zero and a Polarization Index equal to one.

The economy is populated by two types of agents:

1. Firms ($i = 1, \ldots, N_F$), characterised by three state variables: a capital stock $K_i(t)$, an automation level fixed at $A_i = 0$ throughout the baseline, and a roster of connected employees $\mathcal{E}_i$. All firms are initialised with identical capital $K_i(0) = K_0$, ensuring homogeneity on the supply side.
2. Workers ($j = 1, \ldots, N_W$), characterised by their demographic category $\sigma_j \in \{\text{Junior, Senior}\}$, drawn at initialisation with probability $P(\sigma_j = \text{Senior}) = \phi$. Each worker holds a savings stock $S_j(t)$, a skills score h_j, and an employment status $\texttt{status}_j \in \{\texttt{Employed}, \texttt{Unemployed}, \texttt{Retraining}\}$. In the baseline, all workers are initialised with identical savings $S_j(0) = S_0$, skills are fixed at $h_j = 1$, and employment status is permanently set to `Employed`.

The baseline assumptions of equal savings and equal wages ensure that the Gini coefficient and Polarization Index begin at their neutral values, so that any divergence observed in subsequent extensions of the model will be attributable exclusively to differential AI exposure. Unlike the standard Boltzmann wealth model, in which inequality arises endogenously from stochastic bilateral exchange among initially identical agents, our baseline is constructed so that equal initial endowments constitute a stable fixed point of the system dynamics. With uniform wages, a common savings rate, and no random redistribution between workers, the savings of all agents grow in perfect lockstep at every step. The Gini coefficient remains at zero and the Polarization Index at unity throughout the simulation. This stability is not a result but a design requirement: it ensures that the counterfactual is clean, and that any inequality observed in subsequent phases is attributable exclusively to the mechanisms introduced by the AI shock rather than to background stochastic dynamics.

The baseline economy is organised as a static bipartite labour graph $\mathcal{G} = (\mathcal{F} \cup \mathcal{W}, \mathcal{L})$, where $\mathcal{F}$ is the set of firm nodes, $\mathcal{W}$ the set of worker nodes, and $\mathcal{L}$ the set of employment links. Each worker j is assigned to exactly one firm at initialisation through a random matching algorithm: firm $f(j)$ is drawn uniformly from $\mathcal{F}$, and the firm's employee roster is updated accordingly, $\mathcal{E}_{f(j)} \leftarrow \mathcal{E}_{f(j)} \cup \{j\}$. Under full employment, every worker node has exactly one active labour link. The economy therefore operates with no unemployment by construction.

At each simulation step t, the following sequence of events takes place:

1. Production. Each firm i receives a fixed exogenous revenue R, updating its capital:

$$K_i(t) = K_i(t-1) + R. \tag{1}$$

2. Wage distribution. Each firm distributes a uniform baseline wage w_0 to each worker in its roster, reducing its capital accordingly:

$$K_i(t) \;\leftarrow\; K_i(t) - w_0 \cdot |\mathcal{E}_i|. \tag{2}$$

3. Savings and consumption. Upon receiving the wage, each employed worker j saves a fixed fraction $s \in (0,1)$ of their income and returns the remaining to their employer in the form of consumption expenditure:

$$S_j(t) = S_j(t-1) + s \cdot w_0, \tag{3}$$
$$K_i(t) \;\leftarrow\; K_i(t) + (1-s) \cdot w_0. \tag{4}$$

The consumption loop in equation (4) flows directly back to the employing firm, closing the monetary circuit. This accounting closure ensures that no money is created or destroyed within the system at any step, and that the aggregate capital stock of the economy is conserved. We retain this closed-circuit assumption in all subsequent phases, extending it to cover the subsistence expenditures of unemployed workers in the next section.

Two statistics are recorded at the end of each step:

- The Gini coefficient, used to measure income inequality over the distribution of worker savings. The Gini coefficient at time t, $G(t)$, is given by

$$G(t) = \frac{\sum_{j=1}^{N_W} \sum_{k=1}^{N_W} \left| S_j(t) - S_k(t) \right|}{2\, N_W^2\, \bar{S}(t)} \tag{5}$$

  where $\bar{S}(t) = \frac{1}{N_W} \sum_{j=1}^{N_W} S_j(t)$ is mean worker savings at time t. A value of $G(t) = 0$ indicates perfect equality, while values approaching 1 indicate complete concentration.
- The Polarization Index: to account for the structural divergence between demographic groups, we define the Polarization Index as the ratio of mean Senior to mean Junior worker savings:

$$P(t) = \frac{\mu_t^S}{\mu_t^J}, \tag{6}$$

  where μ_t^S and μ_t^J denote mean savings of Senior and Junior workers at time t respectively. By construction $P(t) = 1.0$ when both groups hold identical savings, and $P(t) > 1$ when Seniors accumulate more than Juniors.

By construction, $P(t) = 1$ in the baseline, where all workers earn the same wage, accumulate identical savings, and face zero unemployment.

Prior to introducing technological shocks, the model's behavioral consistency is validated under a static control scenario. Under these benchmark conditions, wealth distribution remains

uniform ($Gini \approx 0$) and labor market polarization is constant ($Polarization = 1$). This flat control confirms that the underlying economy is stable and balanced, ensuring that any subsequent distributional drift can be uniquely attributed to the arrival of AI.

2.2. Extended Model with Stochastic AI Adoption and Technological Unemployment

In this section, we extend the baseline framework by incorporating the technological shock, that is, the adoption of AI by firms. Relative to the baseline, this extension introduces four new mechanisms: a stochastic Poisson adoption process for AI at the firm level; a learning-by-doing productivity channel that raises firm revenue following AI adoption; an asymmetric firing rule that displaces workers upon AI adoption with probabilities that differ by seniority;[1] and a retraining pathway through which displaced workers may recover their labor market position, subject to a savings constraint that generates a poverty trap. Workers who cannot afford to retrain are immediately re-absorbed by the labor market but at a wage penalty, capturing the underemployment dynamics documented in the empirical literature.

This extension augments the state variables of both agent types relative to the baseline:

- Firm agents acquire three additional state variables: a boolean adoption flag $\texttt{ai_unlocked} \in \{\texttt{True}, \texttt{False}\}$, which switches irreversibly to `True` upon AI adoption; a productivity coefficient $\pi_i(t) \in [1, \pi_{\max}]$, initialised at $\pi_i(0) = 1$ and evolving through learning-by-doing after adoption; and a patience counter $\texttt{pat}_i(t) \in \mathbb{N}$, which accumulates during unsuccessful recruitment attempts and governs the firm's reservation skill threshold.
- Worker agents acquire four additional state variables: a skills score $h_j(t) \in [1, \infty)$, initialised at $h_j(0) = 1$ for all workers and increasing upon completion of retraining; a retraining timer $\tau_j(t) \in \mathbb{N}$, which counts down the remaining steps of an active retraining spell; a wage-penalty flag $\texttt{hwp}_j \in \{\texttt{True}, \texttt{False}\}$, set to `True` when a worker is re-hired without having completed retraining; and an individual saving rate $s_j(t)$, which replaces the fixed baseline rate and responds to the worker's current wage relative to the economy-wide baseline:

$$s_j(t) = \text{clip}\left(\bar{s} + \beta_s \left(\frac{w_j(t)}{w_0} - 1\right),\ s_{\min},\ s_{\max}\right), \tag{7}$$

  so that $S_j(t) = S_j(t-1) + s_j(t) \cdot w_j(t)$, with the residual share returned to the firm's capital as consumption. This buffer-stock specification ensures that workers earning above the baseline accumulate faster, while those earning below it save less, consistent with the positive saving-rate–income gradient documented in Dynan et al. (2004).

At each simulation step t, every firm i for which $\texttt{ai_unlocked}_i = \texttt{False}$ draws an independent adoption event. The probability of adoption follows a Poisson arrival process with rate parameter λ_0:

$$P(\text{adopt}_i, t) = 1 - e^{-\lambda_0}. \tag{8}$$

The adoption flag switches irreversibly upon arrival: once a firm adopts AI, it remains automated for the remainder of the simulation. Because draws are independent across firms and across time, adoption diffuses gradually and stochastically through the firm population, generating a realistic pattern of staggered technological entry rather than a simultaneous industry-wide

[1] This key feature of the model is directly related to the empirical findings from Brynjolfsson et al. (2025a).

shock.

Prior to adoption, firm i generates a fixed revenue R_0 at every step, as in the baseline model. Following adoption, firm revenue becomes endogenous, reflecting two channels: a time-varying productivity coefficient $\pi_i(t)$ that captures learning-by-doing gains, and the mean skill level of the firm's current workforce $\bar{h}_i(t)$, which captures the skill-complementarity of AI. Crucially, this specification implies that AI does not act as a mere labor-substitute; rather, its productive potential is unlocked only when combined with high-skill human capital, coherently with economic literature stressing the productivity-boosting role of AI when used as a complement of qualified human labour. Consequently, firms with a more qualified workforce experience endogenous productivity dividends, as skilled workers and AI technologies mutually reinforce each other's efficiency. Net firm revenue after adoption is given by:

$$R_i(t) = \begin{cases} R_0 \cdot \pi_i(t) \cdot \bar{h}_i(t) - C_A & \text{if } R_0 \cdot \pi_i(t) \cdot \bar{h}_i(t) > C_A, \\ 0 & \text{otherwise,} \end{cases} \tag{9}$$

where C_A is a fixed per-step operating cost of AI and we impose the constraint that revenue cannot fall below zero. The mean workforce skill is:

$$\bar{h}_i(t) = \frac{1}{|\mathcal{E}_i(t)|} \sum_{j \in \mathcal{E}_i(t)} h_j(t), \tag{10}$$

with $\bar{h}_i(t) = 1$ if the firm's roster is empty, so that AI continues to generate revenue even without human workers.

Productivity $\pi_i(t)$ evolves through a learning-by-doing process that asymptotically approaches an upper ceiling $\pi_{\max}$:

$$\pi_i(t) = \pi_i(t-1) + \rho \cdot \big(\pi_{\max} - \pi_i(t-1)\big), \tag{11}$$

where $\rho \in (0, 1)$ governs the speed of institutional learning. This specification captures the empirical regularity that productivity gains from general-purpose technologies materialise gradually rather than instantaneously (Brynjolfsson et al., 2021).

Wages are determined by a profit-sharing rule that links worker compensation to firm-level performance. When firm revenue exceeds the pre-adoption baseline, a fraction θ of the surplus is distributed to workers through a wage markup:

$$w_i(t) = \begin{cases} w_0 \left(1 + \theta \, \dfrac{R_i(t) - R_0}{R_0}\right) & \text{if } R_i(t) > R_0, \\ w_0 & \text{otherwise,} \end{cases} \tag{12}$$

where w_0 is the baseline floor wage and $\theta \in [0, 1]$ is the profit-sharing intensity. This specification is consistent with the empirical finding of Brynjolfsson et al. (2025a) that short-run labour market adjustment to AI adoption occurs primarily through employment rather than wage shifts. Workers re-hired without having completed retraining, instead, receive a penalised wage:

$$\tilde{w}_j(t) = \psi \cdot w_i(t), \qquad \psi \in (0, 1), \tag{13}$$

where ψ is an underemployment penalty parameter, reflecting the lower marginal product of workers who have not acquired AI-complementary skills.

The adoption of AI triggers a structural restructuring at the adopting firm, executed in the same step as adoption. Each worker j in the firm's roster is independently evaluated for displacement with a probability that depends exclusively on seniority:

$$P(\text{fired}_j) = \begin{cases} p_J & \text{if } \sigma_j = \text{Junior}, \\ p_S & \text{if } \sigma_j = \text{Senior}, \end{cases} \tag{14}$$

with $p_J > p_S$. This asymmetry reflects the empirical finding from Brynjolfsson et al. (2025a) that early-career workers are disproportionately displaced by AI adoption, as their tasks rely more heavily on codifiable knowledge that AI can substitute, whereas experienced workers are partially protected by tacit knowledge accumulated over their careers. Displaced workers have their employment links severed, their status set to `Unemployed`, and their employer reference cleared. The firing rule executes exactly once per firm, at the moment of adoption, and wages are distributed to all workers before displacement occurs, ensuring that each worker is compensated for their final active step.

Unemployed workers face a binary choice in the first step following displacement, governed by a savings constraint. Let Θ denote the minimum savings required to afford a full retraining cycle, defined as:

$$\Theta = N_{\text{train}} \cdot C_{\text{sub}} + F + w_0 \cdot B_{\text{months}}, \tag{15}$$

where N_{train} is the duration of retraining in steps, C_{sub} is the per-step subsistence cost paid by all non-employed workers, F is a one-time fixed tuition or registration fee that is paid by workers to enroll in the re-training program and $B_{\text{months}} = 3$ is a liquidity buffer representing three additional months of wage income required above the bare training cost, ensuring workers retain a minimal financial cushion post-retraining.

At the first step following displacement, each worker j evaluates once whether their accumulated savings cover the retraining threshold Θ.

1. If $S_j(t_{\text{disp}}) \geq \Theta$ at the moment of displacement, the worker enters `Retraining` status and sets $\tau_j = N_{\text{train}}$. Upon entry, the upfront tuition fee F is immediately deducted from savings. At each subsequent step, the retraining timer decrements and the worker pays C_{sub} from savings. Upon completion ($\tau_j = 0$), skills increase by Δh:

$$h_j \leftarrow h_j + \Delta h, \tag{16}$$

   the worker returns to `Unemployed` status with enhanced human capital.
2. If $S_j(t_{\text{disp}}) < \Theta$, the worker enters the job search pool immediately. If hired, the wage-penalty flag $\texttt{hwp}_j$ is set to `True`, and the worker receives the penalised wage $\tilde{w}_j(t)$ defined in equation (13) for the duration of their employment spell.

This mechanism generates a persistent poverty trap: Junior workers, who face higher displacement probabilities and lower initial savings buffers relative to the retraining threshold, are systematically pushed into the second path, re-entering the labour market at a wage penalty

that further limits their ability to accumulate savings for future retraining. The trap is self-reinforcing, as the wage penalty reduces the speed of savings accumulation, making escape increasingly difficult over time.

Automated firms re-enter the labour market to restore their workforce through a bounded-rationality recruitment process. Hiring is subject to three sequential filters:

1. *Technology filter.* Only firms with $\texttt{ai_unlocked}_i = \texttt{True}$ actively recruit, as their revenue depends on workforce skill composition.
2. *Affordability filter.* Hiring is approved only if the firm maintains a sufficient capital buffer:

$$K_i(t) \geq w_0 \cdot B, \tag{17}$$

   where B is the hiring buffer multiplier.
3. *Capacity filter.* The firm does not hire beyond its initial workforce size: $|\mathcal{E}_i(t)| < |\mathcal{E}_i(0)|$.

If all filters are satisfied, the firm draws a candidate from the unemployment pool using a skill-weighted stochastic draw, so that workers with higher skills have double probability to be selected. The candidate is hired if and only if their skills meet the firm's current reservation skill threshold:

$$S_{\text{res}}(t) = S_{\text{des}} - \texttt{pat}_i(t) \cdot \delta, \tag{18}$$

where S_{des} is the firm's desired skill level, $\texttt{pat}_i(t)$ is the patience counter, and δ is the patience-decay parameter. If the candidate meets the threshold, they are hired and the patience counter resets to zero. If the candidate fails the threshold, the firm remains understaffed and the patience counter increments by one, lowering the reservation skill for the following step. This mechanism generates realistic search frictions: in the short run, firms hold out for skilled workers, prolonging Junior unemployment; in the long run, the eroding threshold forces firms to accept less qualified candidates, gradually stabilising the system.

To preserve causal consistency, the step sequence in this extension follows a two-phase schedule: all firm agents execute their full step routine first - adoption, revenue, wage distribution, firing, and hiring - before any worker agents update their status. The closed monetary circuit introduced in the baseline is extended to cover unemployed workers: subsistence costs C_{sub} paid by non-employed workers are redistributed equally across all firms, preserving aggregate capital conservation throughout all phases of the simulation. Subsistence consumption is capped at available savings to rule out borrowing: a worker whose savings fall to zero incurs no further subsistence charge, preventing negative wealth.

2.3. Extended model with firm-level social pressure for AI adoption and technological unemployment

In the model specification of Section 2.2, AI adoption follows a pure Poisson process: each not-yet-automated firm draws automation independently at a constant per-step hazard λ_0. We now enrich this channel with an endogenous imitation force.

For a firm that has not yet adopted, the arrival rate at step t becomes

$$\lambda_i(t) = \lambda_0 + \gamma\, F(t), \qquad P_i(t) = 1 - e^{-\lambda_i(t)}, \tag{19}$$

where $F(t)$ is the share of firms with `ai_unlocked` = `True` at the beginning of the step. The economic interpretation of the second term, $\gamma F(t)$, is competitive pressure: as more rivals automate and gain a cost and productivity edge, the incentive for a lagging firm to adopt technology rises, so its adoption hazard increases with the share of firms that have already adopted. This is the mean-field limit of Axelrod's social-pressure dynamic: in a globalised economy the firm-to-firm network is complete, so the fraction of adopting neighbours coincides with the global adoption share and no explicit topology need be specified. While with the standard Axelrod model a topological network space where neighbours are defined would be needed, we here assume that all firms are neighbours.[2] Under this limit the mechanism is exactly the diffusion model of Bass (1969), in which λ_0 is the coefficient of innovation (exogenous, generating the pioneers) and γ the coefficient of imitation (endogenous competitive pressure, driving the S-shaped take-off of the adoption curve).[3] Setting $\gamma = 0$ recovers the pure-Poisson specification exactly, preserving backward compatibility and isolating the marginal effect of imitation through a clean $\gamma = 0$ versus $\gamma > 0$ comparison.

$F(t)$ is evaluated once per step, before any firm acts, so that all firms face the same imitation pressure within a step. The shuffled activation order therefore cannot generate within-step cascades, and the dynamics depend only on the aggregate state carried across steps.

3. Parameter Calibration

In order to run simulations and study the model's properties and outcomes, we first need to calibrate its parameters: the present section addresses the issues of parameter calibration and specifies how calibration choices were carried out. In Table 1 all the parameters that are set through calibration, along with the identified values, are listed. Calibration was carried out in sequential stages, ordered by the degree of model involvement required and by the logical dependencies between parameters; to ease the exposition and improve readability, however, in the present section we will discuss the parameters following the order and the thematic subgroups of Table 1.

3.1. System dimensions and initial endowments

The system dimensions and initial endowments are set by normalisation, rather than empirical calibration, with the exception of the Senior worker share ϕ.

The number of firms ($N_F = 100$) and workers ($N_W = 1{,}000$) are chosen to yield a firm-to-worker ratio of 1:10, implying an average initial workforce of ten employees per firm. Results are robust to rescaling, since the model's inequality and polarization dynamics are driven by relative rather than absolute agent counts.

[2] This is a quite realistic assumption in the context of a single-sector globalised economy, under the model's assumptions.

[3] The imitation term is what produces the characteristic S-shape. While the pool of adopters is small, the endogenous force $\gamma F(t)$ is negligible and diffusion is slow; as $F(t)$ grows, each new adopter raises the hazard faced by all remaining firms, creating a positive feedback that accelerates adoption (the convex take-off phase). This acceleration eventually reverses as the residual pool of non-adopters $1 - F(t)$ shrinks toward zero, yielding the concave saturation phase, with an inflection point in between. When $\gamma = 0$ this feedback vanishes and the curve collapses to a concave exponential with no take-off, adoption being concentrated in the earliest steps.

Symbol	Description	Value	Cat.
System dimensions and initial endowments			
N_F	Number of firms	100	N
N_W	Number of workers	1,000	N
K_0	Initial capital endowment per firm	1,000	N
S_0	Initial savings endowment per worker	0	N
ϕ	Share of Senior workers in the labour force	0.30	E
Labour market and consumption			
w_0	Baseline floor wage	20.0	N
$\bar{s}$	Baseline saving rate (at $w_i = w_0$)	0.20	E
β_s	Sensitivity of saving rate to relative wage	0.058	E
$s_{\min}$	Lower bound on individual saving rate	0.00	E
$s_{\max}$	Upper bound on individual saving rate	0.25	E
C_{sub}	Per-step subsistence cost for non-employed workers	10.0	N
θ	Profit-sharing wage markup intensity	0.35	E
ψ	Underemployment wage penalty multiplier	0.85	C
AI adoption and productivity			
λ_0	Poisson arrival rate of AI adoption	0.0101	E
γ	Imitation parameter (competitive pressure)	0.032	E
p_J	Displacement prob. at adoption - Junior	0.60	C
p_S	Displacement prob. at adoption - Senior	0.02	C
R_0	Pre-adoption baseline firm revenue	400.0	N
C_A	Fixed per-step AI operating cost	1.22	E
ρ	Learning-by-doing speed parameter	0.017	E
$\pi_{\max}$	Asymptotic productivity ceiling	1.362	E
Labour market frictions and hiring			
B	Capital buffer multiplier required to hire	3.0	N
S_{des}	Firm's desired skill level for recruitment	1.30	N
δ	Patience decay (S_{des} erosion rate)	0.05	N
Retraining			
N_{train}	Duration of a retraining spell in steps	3	E
Δh	Skill increment upon retraining	0.30	E
F	Training cost	60	E
B_{months}	Savings buffer required above training cost (in months of wage)	3	N
Θ	Minimum savings threshold for retraining	150	D

N = normalisation; **E** = calibrated from empirical sources; **C** = computationally calibrated; **D** = derived.

Table 1: Model parameters, calibration values, and calibration category. Parameters labelled N are fixed by normalisation or internal consistency; those labelled E are pinned to external empirical sources; those labelled C are calibrated computationally to reproduce target properties of the model; Θ is derived analytically as $\Theta = N_{\text{train}} \cdot C_{\text{sub}} + F + w_0 \cdot B_{\text{months}}$.

Initial firm capital ($K_0 = 1{,}000$) is set as a normalisation anchor: at the baseline wage of $w_0 = 20$ and ten workers per firm, a firm covers approximately 2.5 months of wage costs from its initial endowment before revenue arrives, ensuring no firm faces a liquidity crisis in the first steps of the simulation from wage obligations alone.

Initial worker savings are set to zero ($S_0 = 0$), assuming that workers enter the simulation with no pre-accumulated buffer stock. This choice maximises the severity of the poverty trap mechanism: workers displaced early in the adoption wave, before they have had time to accumulate savings from wages, are structurally unable to afford retraining. Setting $S_0 > 0$ would attenuate this mechanism by giving early-displaced workers a partial buffer; we treat $S_0 = 0$ as the baseline and performed robustness checks to confirm that results were qualitatively unchanged for $S_0 > 0$.

The Senior worker share $\phi = 0.30$ was calibrated from external data. Following the Bureau of Labor Statistics Employee Tenure Summary (U.S. Bureau of Labor Statistics, 2024a), approximately 26% of US wage and salary workers held ten or more years of tenure with their current employer in January 2024. We interpret workers with substantial tenure as a proxy for the accumulated tacit knowledge that Brynjolfsson et al. (2025a) identify as the primary buffer against AI-driven displacement, and round to $\phi = 0.30$ as a conservative estimate. The precise value is subject to uncertainty because the model's binary Junior/Senior distinction does not map cleanly onto the continuous tenure distribution in the data.

3.2. Labour market and consumption

The baseline wage $w_0 = 20$ is set by normalisation. We adopt a labour share of 0.50 as a stylised round-number benchmark, broadly in line with—though slightly conservative relative to—long-run estimates of the labour share in the US private sector, which lie in the range of roughly 0.57–0.60 (U.S. Bureau of Labor Statistics, 2024b). This implies that half of each firm's baseline revenue is distributed as wages. With baseline firm revenue $R_0 = 400$ and an average of ten workers per firm ($N_W / N_F = 1{,}000/100 = 10$), the total wage bill per firm per step equals $0.50 \times 400 = 200$, giving $w_0 = 200/10 = 20$ model units. This value also serves as the exchange-rate anchor for converting real-world monetary quantities—re-training program tuition, subsistence costs—into model units, via the implicit mapping $w_0 = 20 \leftrightarrow \$4{,}500$ per month, a representative monthly wage for college-educated workers in AI-exposed occupations in the United States.

The individual saving rate $s_i(t)$ responds to the worker's current wage $w_i(t)$ relative to the baseline w_0:

$$r_i(t) = \frac{w_i(t)}{w_0}, \qquad s_i(t) = \mathrm{clip}\Big(\bar{s} + \beta_s\,(r_i(t) - 1),\ s_{\min},\ s_{\max}\Big), \tag{20}$$

so that $\mathrm{Savings}_i(t) = \mathrm{Savings}_i(t-1) + s_i(t)\,w_i(t)$, with the residual share returned to the firm's capital. In this way, $\mathrm{clip}(x, s_{\min}, s_{\max})$ truncates its argument to the interval $[s_{\min}, s_{\max}]$, returning $s_{\min}$ if $x < s_{\min}$, $s_{\max}$ if $x > s_{\max}$, and x otherwise. This keeps the individual saving rate within an economically admissible range, preventing negative saving (no borrowing) at very low relative wages and implausibly high saving rates at the top. The interval is not arbitrary: its bounds are grounded in Dynan et al. (2004), who document saving rates ranging from about 0%

in the bottom income quintile to roughly 25% in the top.[4]

The level $\bar{s} = 0.20$ is anchored to the aggregate U.S. personal saving rate over 2022–2025 (FRED, series PSAVERT). The sensitivity β_s - how far the individual saving rate departs from $\bar{s}$ as the relative wage varies - is anchored to Dynan et al. (2004), who report the coefficient of a linear regression of the saving rate on income, in percentage points per $10,000 of income. For the PSID *active saving* measure (col. 5 - saving out of current income net of capital gains, the concept closest to our $s_i \times w_i$) the coefficient is 0.015. Converting from the "per $10,000" unit to the model's unit yields (per unit of relative wage r_i)[5]

$$\beta_s = 0.015 \times \frac{38{,}500}{10{,}000} \approx 0.058. \tag{21}$$

This value is consistent, as a robustness check, with an independent estimate obtained by linearly interpolating Dynan et al. (2004)'s quintile saving rates(tab. 3) against the Census Bureau's aggregate income shares by quintile (*Income in the United States*, tab. 2): that interpolation yields $\beta_s \approx 0.048$, of the same order of magnitude.[6]

The bounds $s_{\min}$ and $s_{\max}$ derive from the same source: Dynan et al. (2004) document saving rates ranging from 0% (lowest quintile) to above 25% (highest quintile) of current income. We therefore set $s_{\min} = 0.00$ and $s_{\max} = 0.25$. The asymmetric distance from $\bar{s}$ (0.20 downward, 0.05 upward) faithfully reflects the nonlinearity of the empirical relationship - nearly flat across the lower quintiles, steep only at the top of the distribution (Dynan et al. (2004), fig. 3: the implied slope between the fourth and fifth quintiles is almost 15 times that between the first and second) - and is therefore not an artefact to be corrected.

The per-step subsistence cost C_{sub} paid by unemployed workers is set proportionally to the baseline wage at a ratio of $C_{\text{sub}}/w_0 = 0.50$, giving $C_{\text{sub}} = 10$ model units. This implies that surviving one step of unemployment costs half a monthly wage in living expenses, a plausible order of magnitude for basic consumption expenditure net of discretionary spending. Unemployed workers' subsistence expenditure is routed back to firms in equal shares, preserving the closed monetary circuit of the model.

[4] Although Dynan et al. (2004) rely on relatively dated micro data (PSID, 1984–89), more recent U.S. evidence confirms that the positive saving-rate–income gradient is not an artefact of that period. Using distributional tax records through 2019, Mian et al. (2024) document that the saving rate of the top 1% rose from about 43% to over 54% of disposable income since the 1980s, while it declined for the rest of the distribution - if anything strengthening the gradient over time. We therefore retain Dynan et al. (2004) coefficient, which is the only estimate expressed in the functional form required by our specification (a linear saving-rate–income slope on active saving, net of capital gains).

[5] Dynan et al. (2004)'s coefficient is expressed per $10,000 of income, whereas in the model β_s measures the response to a unit change in the relative wage r_i, i.e. to a doubling of the wage. Since the mean income in the underlying PSID sample is $\approx$ $38,500, a doubling ($r_i$ from 1 to 2) corresponds to an income increase of $\approx$ $38,500, that is 3.85 times the $10,000 to which the coefficient refers. Hence $\beta_s = 0.015 \times 3.85 \approx 0.058$: if each additional $10k of income raises the saving rate by 0.015, then $38.5k raises it by roughly 0.058.

[6] $\bar{s}$ and β_s come from two different sources: a contemporaneous macro aggregate (PSAVERT) for the level, a historical micro estimate (Dynan et al., PSID 1984–89) for the sensitivity. Anchoring $\bar{s}$ to Dynan as well would yield a level too low (≈ 0.06) for workers to ever accumulate the retraining threshold Θ, collapsing the model into a near-universal poverty trap. We therefore anchor the level to the current macro figure and only the sensitivity to Dynan, on the assumption that behavioural sensitivity to relative income is more stable over time than the aggregate saving rate, which moves with the business cycle.

Regarding the calibration of the profit-sharing wage markup intensity θ, wages in the model are not fixed: a firm that raises its revenue above the pre-adoption baseline R_0 passes a fraction of the surplus on to its workforce, according to

$$w_{i,t} = w_0 \left(1 + \theta \cdot \frac{R_{j,t} - R_0}{R_0}\right), \tag{22}$$

so that θ governs how strongly firm-level gains translate into wages. Conceptually, θ is thus a rent-sharing parameter: it captures how much of an increase in a firm's profitability is passed through to worker pay, independently of what caused that increase. This is a structural feature of the firm-worker relationship - bargaining power, labour-market norms, institutions - rather than a property of the specific shock, which is why an estimate identified in one setting can reasonably discipline its value in the present setting.

Kline et al. (2019) link the universe of U.S. patent applications[7] to firm and worker tax records[8] and identify the pass-through of firm surplus to wages by comparing firms whose applications were initially allowed with those initially rejected. The patent grant is only their identification device - a source of exogenous variation in firm profitability - and not a feature we import into our setting. The firms in the model do not patent, but the rent-sharing rate estimated by Kline et al. (2019) is a general measure of how firms share gains with workers, applicable whatever the underlying source of the gain (here, AI-driven productivity rather than a patent). They estimate an earnings-to-surplus elasticity of approximately 0.35 (a monetary pass-through of about 29 cents per dollar of surplus, averaged across the firm's workers).

We adopt this elasticity as our pass-through intensity, setting $\theta = 0.35$, which to our knowledge is the cleanest causal estimate of firm-to-worker rent-sharing available. Like the rest of this literature, Kline et al. (2019) define pass-through relative to firm surplus, whereas θ scales the relative change in revenue. In principle this matters, because revenue and surplus move together only up to the firm's cost structure, so the same elasticity need not carry over one-to-one across the two bases. In our setting, however, the gap is second-order: the AI cost C_A is small relative to the productivity gains it unlocks, so a firm's net revenue tracks its surplus closely, and the two denominators nearly coincide. We therefore read Kline et al.'s estimate as disciplining the order of magnitude of θ rather than its exact value, and probe the remaining range in the sensitivity analysis.

The wage penalty ψ applied to workers who are rehired without having completed retraining, ψ, has no direct empirical counterpart: its effect on the aggregate wage gap depends on the endogenous composition of workers re-hired without retraining, which only the simulation can produce. We therefore calibrate it via the Simulated Method of Moments (SMM), targeting Fact 5 from Brynjolfsson et al. (2025a), who document that labor market adjustments to AI adoption are visible primarily in employment rather than in compensation: the wage divergence between more-exposed and less-exposed groups is far smaller than the corresponding employment divergence. We translate this qualitative finding into a quantitative target by requiring that

[7] Filed with the U.S. Patent and Trademark Office (USPTO), the federal agency that grants patents.

[8] From the Internal Revenue Service (IRS), the U.S. federal tax authority; these are administrative records covering essentially the full population of U.S. firms and employees, offering far greater coverage and reliability than survey data.

the model-implied wage gap between Senior and Junior workers - the difference in their cumulative wage growth relative to the baseline wage over the 36-step calibration horizon - equal approximately 15% of the already-calibrated employment gap. With $p_J = 0.60$ and $p_S = 0.00$ (whose calibration is discussed in Section 3.3), this employment gap is 16 percentage points, yielding a wage-gap target of 2.4 percentage points.

The profit-sharing markup θ is held fixed at 0.35 throughout this exercise, following its above-mentioned direct calibration. This separation is justified on identification grounds: θ scales the wages of all employed workers proportionally, regardless of seniority, and therefore has a negligible effect on the wage gap between the two groups, which is instead governed almost entirely by ψ - the latter applying asymmetrically to workers re-hired without retraining, who are disproportionately Junior. We accordingly calibrate ψ alone via a one-dimensional grid search, holding θ, p_J, and p_S fixed at their previously calibrated values, over $\psi \in [0.75, 0.95]$[9] with a step size of 0.01 and 15 simulation seeds per value, minimizing

$$L(\psi) = \left(\overline{\text{wage_gap}}(\psi) - 0.024\right)^2 \tag{23}$$

The search identifies

$$\psi = 0.85, \tag{24}$$

with a simulated wage gap of 2.47% (target 2.4%; squared loss $\approx 5.2 \times 10^{-7}$) against a simulated employment gap of 16%, for a wage-gap-to-employment-gap ratio of 0.154 (target 0.15), and a Monte Carlo standard error across seeds of 0.14 percentage points, indicating a precise and stable fit that lies comfortably within the interior of the tested grid.

3.3. AI adoption and productivity

We now move on the calibration of parameters related to the AI adoption process and its productivity implications. Starting with the Poisson arrival rate of AI adoption λ_0 and the imitation parameter γ, we know that they are exogenous to labour market dynamics, so that the pair (λ_0, γ) can be calibrated on its own, using only the adoption trajectory $F(t)$. A single moment, however, constrains only a locus of (λ_0, γ) pairs and cannot separate the exogenous channel (λ_0, which generates the first AI pioneers) from the imitative one (γ, which drives the endogenous adoption take-off). We therefore fix γ from external evidence and calibrate only λ_0 on an empirical moment.

For γ, we draw on the meta-analysis of Bass-model estimates by Sultan et al. (1990), who report a mean imitation coefficient $q \approx 0.38$ on annual data. Since one step corresponds to one month in the model, a first-order conversion (linear approximation) gives $\gamma = q/12 \approx 0.032$. This is a deliberately conservative use of the literature: rather than estimating the strength of imitation from our own adoption data, we import a value established across many diffusion studies, leaving only the exogenous rate to be pinned down internally.

[9] The search range $\psi \in [0.75, 0.95]$ is chosen to span the interval of economic plausibility suggested by the external literature on displaced-worker and occupation-mismatch wage penalties: its bounds correspond, respectively, to a wage penalty of about 25% (Couch and Placzek, 2010, initial displacement loss) and about 5% (Krolikowski, 2023, tight-labor-market reemployment), with the SMM procedure selecting the point within this range that best matches the Fact 5 wage-gap moment.

For λ_0 we target the period-36 adoption share reported by Brynjolfsson et al. (2025a), $F(36) = 0.46$, the empirical window used throughout the calibration (October 2022 to September 2025). We recover λ_0 by inverting the closed-form Bass adoption curve,

$$F(T) = \frac{1 - e^{-(\lambda_0+\gamma)T}}{1 + (\gamma/\lambda_0)\, e^{-(\lambda_0+\gamma)T}} \tag{25}$$

solving $F(36) = 0.46$ for λ_0 via Brent's root-finding method (`scipy.optimize.brentq`), which yields

$$\lambda_0 = 0.0101, \qquad \gamma = 0.032. \tag{26}$$

As an external check, the Real-Time Population Survey (RPS) of Bick et al. (2024) reports that 26.5% of employed U.S. workers used generative AI at work by late 2024 (step 22 of our horizon); the calibrated Bass curve implies $F(22) \approx 0.27$, closely matching this independent moment.[10] The recursion is the deterministic, expected-value counterpart of the model's stochastic adoption process: it replaces each firm's random draw with its expected adoption probability applied to the residual pool of non-adopters, which makes the calibration a one-dimensional root-finding problem, fully reproducible from equation (25).

We then move to the calibration of the displacement probabilities for Junior workers (p_J) and Senior workers (p_S). At the moment of adoption, indeed, each worker is displaced with a probability p_J or p_S that depends only on seniority. Unlike parameters that can be derived directly from a formula (such as λ_0), p_J and p_S have no closed-form mapping to an observable moment: their effect on cohort employment emerges only through the model's full dynamics of displacement, retraining, and re-hiring. We therefore recover them through a SMM approach: for each candidate pair we run the model over the calibration horizon, average the resulting moments across seeds to integrate out Monte Carlo noise, and select the pair minimising the squared distance to the empirical targets. The targets are the two central facts of Brynjolfsson et al. (2025a), measured under their most conservative specification (with firm-time fixed effects): a 16% relative decline in the employment of early-career workers (ages 22-25) in the most AI-exposed occupations, and no AI-driven decline for experienced workers. In the model these map to

$$\Delta_{\text{Junior}} = -0.16, \qquad \Delta_{\text{Senior}} = 0.00, \tag{27}$$

where Δ is the percentage change in each cohort's employment between $t = 0$ (October 2022) and $t = 36$ (September 2025, the end of the empirical window). We minimise the equally weighted loss

$$L(p_J, p_S) = \left(\overline{\Delta}_{\text{Junior}}(p_J, p_S) + 0.16\right)^2 + \left(\overline{\Delta}_{\text{Senior}}(p_J, p_S)\right)^2, \tag{28}$$

where $\overline{\Delta}$ is the seed-averaged simulated moment (15 seeds per pair), holding all other parameters at their calibrated values. The grid covers $p_J \in \{0.60, 0.65, 0.70, 0.75, 0.80\}$ and $p_S \in \{0.00, 0.02, 0.04, \ldots, 0.14\}$ ($5 \times 8 = 40$ pairs), yielding

$$p_J = 0.60, \qquad p_S = 0.00, \tag{29}$$

with simulated moments $\overline{\Delta}_{\text{Junior}} = -16.0\%$ and $\overline{\Delta}_{\text{Senior}} = +0.0\%$.

[10] This is a proxy check only: the RPS measures worker-level use, whereas $F(t)$ is firm-level adoption in our model, and worker use typically runs ahead of formal firm adoption (Bick et al., 2024).

The calibrated $p_S = 0.00$ sits at the lower end of its grid, but this is not a grid artefact: indeed, even widening the grid, the result is the same. Zero is the natural lower bound of a probability, and the value carries a clear economic reading - AI adoption displaces no experienced workers, consistent with the stable senior employment documented by Brynjolfsson et al. (2025a). Because $p_S = 0.00$, Seniors never enter unemployment and are therefore never subject to the hiring-friction mechanism of Section 3.3. They are instead affected only indirectly, through firm-level workforce composition: as displaced Juniors are gradually replaced by a smaller or less-skilled pool of employees, the mean workforce skill $\bar{h}_i(t)$ (Equation 10) that enters firm revenue can shift, which in turn affects the profit-sharing wage markup (Equation 12) paid to all employees, Seniors included. For numerical robustness, the production simulations reported in Section 4.2 and Table 1 use $p_S = 0.02$ rather than the boundary-optimal $p_S = 0.00$ identified above. The two values are practically indistinguishable in their labour market implications ($\Delta_{\text{Senior}} \approx -0.6\%$ under $p_S = 0.02$, against the targeted 0.0%), and we verified that all qualitative results reported below are unaffected by this choice.

The calibration of p_J and p_S was carried out on the Axelrod/Bass specification ($\gamma = 0.032$); the pure-Poisson specification ($\gamma = 0$) inherits p_J and p_S as a counterfactual and was not separately recalibrated.

The pre-adoption baseline firm revenue ($R_0 = 400$) is a scale normalisation that anchors the monetary units of the model. Its absolute level is irrelevant for the purposes of any economic interpretation, and it is set ensuring internal consistency with the other monetary parameters.

Once a firm has adopted AI, it incurs a fixed AI operating cost C_A in every subsequent step that is subtracted from its gross revenue: net revenue for firm j is thus $R_{j,t} = R_0 \cdot \pi_i(t) - C_A$ for adopters, and R_0 otherwise. We calibrate the ratio of recurring AI cost to firm revenue, which is the scale-free quantity the dynamics depend on.[11] Enterprise subscriptions over 2023-2024 - Microsoft 365 Copilot (launched November 2023 at $30 per user per month) and ChatGPT Team/Business ($25-30 per user per month) - imply a representative price of about $27.5 per user per month, or roughly $275 per month for a representative firm. Converting to model units with the same exchange rate $k = 225$ used for the training cost F,[12] this places the recurring AI cost at

$$C_A = \frac{27.5 \times 10}{225} \approx 1.22, \tag{30}$$

about 0.3% of baseline revenue $R_0 = 400$. Because C_A scales with the firm together with R_0, this ratio - not the nominal value - is what the calibration fixes.

Note that we model only the recurring cost C_A and omit the substantial one-off setup cost (integration, training, governance) that should represent the larger component of AI adoption costs. This is a deliberate scoping choice, justified on three grounds. First, we study large firms, for which both costs are of second order relative to revenue. Second, firms are ex-ante identical

[11] The model is defined only up to a scale normalisation: the nominal levels (ten workers per firm, $R_0 = 400$, $K_0 = 1{,}000$, $w_0 = 20$) are fixed for internal consistency, not as absolute magnitudes, and can be rescaled jointly - across all monetary quantities - without affecting the model's outputs, which are ratios and distributional indices (the Gini coefficient, the Polarization Index, displacement and re-skilling shares) invariant to such rescaling. A firm of "ten workers" is thus a scaled representative of a firm of arbitrary size.

[12] k is the ratio between a representative monthly wage for college-educated workers in AI-exposed occupations ($\approx$ \$4,500) and the model's baseline wage $w_0 = 20$, so that $k = 4{,}500/20 = 225$.

and all eventually adopt under the Poisson process, so a common entry cost would shift every firm equally and leave the labour-market dynamics - our object of study - unchanged. Third, being paid once, such a cost is quickly amortised: a setup cost of the magnitude implied by the Atlanta Fed data (≈ 90 in model units) would be sizeable at adoption (about 22% of a firm's per-step revenue) yet fall below 2% of cumulative revenue within a year and become negligible over the full horizon. Consistent with our focus - not firms' investment decisions but the redistributive consequences of adoption - we fold the one-off investment into the normalised initial conditions and retain only C_A. We nonetheless note that this abstracts from the transition costs firms face at the moment of adoption; introducing a separate one-off cost is a natural extension we leave to future work.

Finally, we move to the learning parameters ρ and $\pi_{\max}$, which are jointly identified through a two-moment calibration that is fully independent of the rest of the model and can be conducted analytically. The closed-form solution to the productivity accumulation Equation 11 is

$$\pi_i(t) = \pi_{\max} - (\pi_{\max} - 1)(1 - \rho)^t \tag{31}$$

and we match it to two empirical moments simultaneously, yielding a system of two equations in two unknowns that is generically identified.

The first empirical moment requires that firm productivity reaches 90% of its asymptotic ceiling by simulation step 60, corresponding to a five-year horizon. This target is grounded in Brynjolfsson et al. (2021), who document that productivity gains from general-purpose technologies typically materialise over 5-10 years due to the time required to accumulate complementary organisational capital and reconfigure firm-level routines:

$$\pi_i(60) = 0.90 \cdot \pi_{\max} \tag{32}$$

The second moment anchors the level of productivity at an intermediate horizon. Brynjolfsson et al. (2025b) document that access to a generative AI assistant raised worker throughput by approximately 14% in a large-scale field study, providing an empirical estimate of the short-run productivity gain from AI adoption at $t = 30$ steps (two and a half years post-adoption):

$$\pi_i(30) = 1.14 \tag{33}$$

Substituting the analytical expression for $(1 - \rho)^{30}$ implied by the first moment into the second reduces the system to a single equation in $\pi_{\max}$, solved numerically. The identified parameter values are $\pi_{\max} = 1.362$ and $\rho = 0.0162$, implying that AI adoption raises long-run firm productivity by approximately 36% above the pre-adoption baseline, with 90% of that gain materialising within five years. Both moments are reproduced exactly by the calibrated pair, as verified by direct substitution into the closed-form trajectory. For implementation we round to $\rho = 0.017$.

3.4. Labour market frictions and hiring

The capital buffer multiplier $B = 3.0$ governs the minimum capital reserve a firm must hold before it is permitted to hire: a firm initiates recruitment only if $K_i(t) \geq B \cdot w_0$. The value $B = 3.0$ implies that a firm must hold at least three months of wage costs in reserve before hiring,

a conservative liquidity requirement consistent with standard corporate cash management practice. A diagnostic grid search varying B across the range $[1.0, 5.0]$ confirmed that this parameter is non-binding in practice. B is therefore retained as a structural safeguard against degenerate balance-sheet dynamics rather than as an active determinant of labour market outcomes, and its value is set by convention.

The desired skill threshold S_{des} and the patience decay parameter δ govern the firm's hiring friction according to Equation 18. In particular, S_{des} represents the initial reservation skill level that a candidate must hold to be hired immediately while δ controls how quickly that threshold erodes over time with each unsuccessful search attempt. Unlike the parameters calibrated against external empirical moments, these two parameters describe an internal firm decision rule - the dynamic adjustment of hiring standards under labour market frictions - for which no directly observable real-world counterpart exists in available datasets, at least to our knowledge. We thus perform a sensitivity analysis keeping all other parameters fixed in order to ensure that a theoretically motivated choice for the values of S_{des} and δ does not influence the main results of the model. To do so, we carry out a systematic grid search over $S_{\text{des}} \in \{1.10, 1.15, 1.20, 1.25, 1.30, 1.35, 1.40\}$ and $\delta \in \{0.02, 0.06, 0.10, 0.14, 0.18\}$, confirming that both parameters leave the model's qualitative findings invariant: in all 35 combinations tested across 10 independent replications each, the Gini coefficient rises meaningfully above zero and the Polarization Index exceeds unity following AI adoption. We therefore set $S_{\text{des}} = 1.30$ and $\delta = 0.05$ on theoretical grounds. The value $S_{\text{des}} = 1.30$ is chosen to equal exactly $1 + \Delta h$, the skill level a worker reaches upon completing retraining, so that successful retraining fully resolves the firm's skill concern and grants immediate re-employment (see Section 3.5 for the calibration of Δh). The value $\delta = 0.05$ implies that a firm lowers its reservation skill by 0.05 per step of unsuccessful search, reaching the non-retrained worker's skill floor ($h = 1.0$) after six failed attempts.

3.5. Retraining

The duration of a retraining spell N_{train} is calibrated empirically from data on the duration of technology bootcamp programmes held in the U.S., the closest real-world analogue to the retraining pathway modelled here.[13] According to Course Report, an independent organisation that has tracked the coding bootcamp industry since 2013, full-time programmes average approximately 14 weeks in length, with a documented range of 12 to 20 weeks (Course Report, 2025). Setting one simulation step equal to one month, $N_{\text{train}} = 3$ corresponds to approximately 13 weeks, consistent with reported values.

The skill gain Δh experienced by a worker upon completing retraining is anchored to evidence on earnings returns to coding bootcamp programmes. A Gallup-2U study of bootcamp graduates finds that workers transitioning from a non-STEM to a STEM role experienced median income growth of approximately 29% (Gallup, Inc., 2024). We interpret this income gain as a proxy for the increase in AI-complementary capability, and set $\Delta h = 0.30$. This value has

[13] In the framework of our model, retraining does not represent a general skill-enhancing program, but a program specifically aimed at the accumulation of AI-complementary skills and capabilities for workers. For this reason we choose to calibrate retraining parameters directly from data on coding bootcamp programs, that we consider as a plausible real-world counterpart of the retraining program that we model.

an additional structural property that is theoretically desirable: since a retrained worker's skill reaches $1.0 + \Delta h = 1.30 = S_{\text{des}}$, retraining fully resolves the firm's skill concern and grants immediate re-employment without requiring any patience erosion. This ensures a clean separation between workers who are re-absorbed promptly upon completing retraining, and unskilled workers, who remain subject to the full hiring friction mechanism.

The one-time tuition cost of retraining, F, is converted from real-world bootcamp fees using the implicit exchange rate between model units and dollars. Setting $w_0 = 20$ model units equal to a representative monthly wage of \$4,500 for college-educated workers in AI-exposed occupations gives a conversion factor of $k = 4{,}500/20 = 225$. Course Report documents an average full-stack boot camp tuition of \$13,584, with a range of \$7,800 to \$21,000 (Course Report, 2024). Applying the conversion factor gives $F = 13{,}584/225 \approx 60$ model units.

Beyond the bare cost of a retraining cycle, we assume workers require a precautionary liquidity cushion B_{months} before committing to retraining, representing the additional months of wage income they must hold to sustain themselves during and immediately after the spell without falling into insolvency. We set $B_{\text{months}} = 3$, i.e. three months of baseline wage income ($w_0 \cdot B_{\text{months}} = 20 \times 3 = 60$ model units), consistent with the standard personal-finance benchmark of maintaining roughly three months of expenses as an emergency buffer. This precautionary component tightens the credit constraint that generates the poverty trap. Its effect on re-skilling and inequality outcomes is examined in the robustness analysis of Appendix A, where some alternative specifications with $B_{\text{months}} = 0$ are tested.

Finally,the minimum retraining threshold Θ is derived analytically from the parameters above as follows:

$$\Theta = N_{\text{train}} \cdot C_{\text{sub}} + F + w_0 \cdot B_{\text{months}} = 3 \times 10 + 60 + 20 \times 3 = 150. \tag{34}$$

Θ represents the minimum savings a worker must have accumulated at the moment of displacement to afford the upfront tuition fee F, the subsistence costs C_{sub} sustained during the three-step retraining spell, and the precautionary buffer $w_0 \cdot B_{\text{months}}$ that guarantees a minimal financial cushion post-retraining.

4. Model properties and simulation results

The model presented in Section 2 is stochastic along three main independent dimensions: the timing of AI adoption, the seniority-driven displacement drawn at each adoption event, and the credential-weighted hiring lottery. The Gini coefficient and the Polarization Index are therefore random trajectories, and any single-seed run would confound the underlying economic mechanism with stochastic variability. We thus implement Monte Carlo simulations to study the model properties and outcomes. Rather than fixing the replication count by convention, however, we first implement a Statistical Model Checking (SMC) procedure to determine the number of runs. Said procedure is discussed in Section 4.1. The resulting number of replications is subsequently used to run all the simulations presented in Section 4.2 and the robustness checks in Appendix A.

4.1. Statistical Model Checking (SMC)

The number of Monte Carlo replications is determined endogenously through a SMC procedure, following the transient-estimation framework of Vandin (2024) and the methodology advocated for agent-based models by Fagiolo et al. (2007). All results in this section concern the Axelrod specification of the model with social imitation, with parameters already set as in the baseline results that we present in the results section ($\gamma = 0.032$, $B_{\text{months}} = 3$, $\delta = 0.05$).

To implement SMC, the model is wrapped in a lightweight adapter exposing three methods in the spirit of the MultiVeStA interface: `set_simulator_for_new_simulation` (which reinitialises the economy with a new random seed), `perform_one_step` (which advances the simulation by one month), and `eval` (which reads a named observable from the model's data collector). The driver runs independent replications sequentially, starting from $N_{\text{init}} = 30$ and adding batches of 20, estimating at each step t the expected trajectory $E[\text{obs}(t)]$ and its t-Student 95% confidence interval. Stopping occurs when the half-width of the confidence interval falls below the target threshold at every step simultaneously for both observables. Because the adoption process is one-way and the retraining decision is irreversible, the model dynamics are transient and non-stationary by construction; steady-state estimation would be ill-posed on this absorbing process, making transient estimation the appropriate choice.

Two observable targets - the Polarization Index and the Gini coefficient - are monitored, and they require different stopping criteria. The Polarization Index P is bounded away from zero throughout the horizon (by construction $P(t) \geq 1$), so a relative half-width target is well-conditioned at every step. We require:

$$\frac{t_{\alpha/2,\,N-1} \cdot \hat{\sigma}_t(P)}{\sqrt{N} \cdot |\hat{\mu}_t(P)|} \leq 0.05 \quad \forall\, t \in \{1, \ldots, 150\}. \tag{35}$$

The Gini coefficient, on the contrary, requires a different approach. As the coefficient starts from zero and then rises gradually over the horizon, a relative target would be ill-conditioned in the early near-zero steps, where the denominator $|\hat{\mu}_t(\text{G})|$ approaches zero and a tiny cross-seed standard deviation would demand an excessively large number of replications to characterise an economically irrelevant warm-up phase. We thus adopt instead an absolute half-width target:

$$\frac{t_{\alpha/2,\,N-1} \cdot \hat{\sigma}_t(G)}{\sqrt{N}} \leq 0.01 \quad \forall\, t \in \{1, \ldots, 150\}. \tag{36}$$

This targets precision on the part of the trajectory where it matters: the absolute criterion binds at the step where cross-simulation dispersion $\hat{\sigma}_t(G)$ is largest, which for the Gini falls near the end of the horizon where inequality is fully developed and the variability across seeds is economically meaningful. The effective sample size is therefore dimensioned on the informative portion of the trajectory, not on the initial transient.

The SMC procedure terminates at $N = 50$ replications under the main model specification. The two observables reach their worst-scenario confidence interval at different steps. For the Gini coefficient, the binding step is $t^*_G = 30$, where the absolute half-width reaches 0.0082 against a target of 0.01 (sample mean $\hat{\mu}_{30}(G) = 0.1241$). For the Polarization Index, the binding

step is $t_P^* = 42$, where the relative half-width is 0.0148, well within the 5% target (sample mean $\hat{\mu}_{42}(P) = 1.3014$). The Gini coefficient is thus the tighter constraint in absolute terms, binding earlier in the transient, while the Polarization Index remains comfortably within target throughout the horizon.

Given the SMC outcome, the number of Monte Carlo runs $N = 50$ is thus applied to all four model specifications that we test, reported respectively in Section 4.2 and Appendix A. The 95% CI guarantee holds in strict terms only for the main specification on which N was dimensioned; for the robustness variants it provides a consistent statistical footing without claiming identical precision.

4.2. Simulation and Model Results

The present Section reports the outcomes of the Monte Carlo simulation of the Axelrod specification of our model of AI adoption, in which each firm's per-period adoption hazard is $\lambda_i(t) = \lambda_0 + \gamma F(t)$, with $\lambda_0 = 0.0101$ the calibrated Bass innovation coefficient, $\gamma = 0.032$ the imitation coefficient, and $F(t)$ the economy-wide share of firms already automated. All results are obtained from running 50 independent Monte Carlo replications over a horizon of $T = 150$ monthly steps, with a labour market of 1,000 workers (30% Senior, 70% Junior) matched to 100 firms. Unless otherwise stated, we report the cross-seed mean trajectory together with the 5th–95th percentile band, so that reported quantities reflect both the central tendency and the seed-to-seed variability of the stochastic process.

We organise the discussion of results around the two dimensions along which the model is most sensitive: the hiring-selectivity parameter δ, which governs how quickly a firm's skill reservation erodes as it fails to hire, and the precautionary savings buffer B_{months}, which enters the endogenous retraining threshold $\Theta = N_{\text{train}} \cdot C_{\text{sub}} + F + w_0 \cdot B_{\text{months}}$. The main specification fixes $B_{\text{months}} = 3$ and $\delta = 0.05$; three robustness variants ($\delta \in \{0.05, 0.02\}$ crossed with $B_{\text{months}} \in \{3, 0\}$) isolate the marginal and joint effects of these two frictions.

This Section is thus structured as follows: we first present the baseline dynamics of inequality, polarization, unemployment and seniority-conditional savings and wages (Figure 1), followed by the reskilling-versus-diffusion dynamics (Figure 2). We then summarise the terminal-step distributions across the 50 seeds (Table 2) and close with a comparative analysis of the four parameter configurations, which identifies ($\delta = 0.02$, $B_{\text{months}} = 3$) as the most adverse regime along almost every dimension and ($\delta = 0.05$, $B_{\text{months}} = 0$) as the most benign.

Three qualitative regularities emerge and are documented in detail below. First, inequality is a transient phenomenon: the Gini coefficient, the Junior-only Gini coefficient, and the Polarization Index all rise sharply during the adoption wave, peak around the middle of the horizon (roughly steps 45–67, coinciding with the diffusion of automation across firms), and then partially recede as displaced workers are progressively rehired. Second, the two frictions act through distinct channels: a lower patience decay ($\delta = 0.02$) keeps firms selective for longer, prolonging unemployment spells and widening the terminal Senior/Junior gap, whereas a higher savings buffer ($B_{\text{months}} = 3$) raises the retraining threshold and thereby depresses the share of displaced workers who successfully reskill. Third, although the direct wage gap between cohorts remains modest at the terminal step ($\sim$ 2–4%), it compounds into a much

larger savings gap (a polarization ratio of 1.17–1.31), illustrating how a small but persistent flow differential amplifies into a sizeable stock disparity.

Figure 1 -- Main spec (buffer_months=3, patience_decay=0.05)

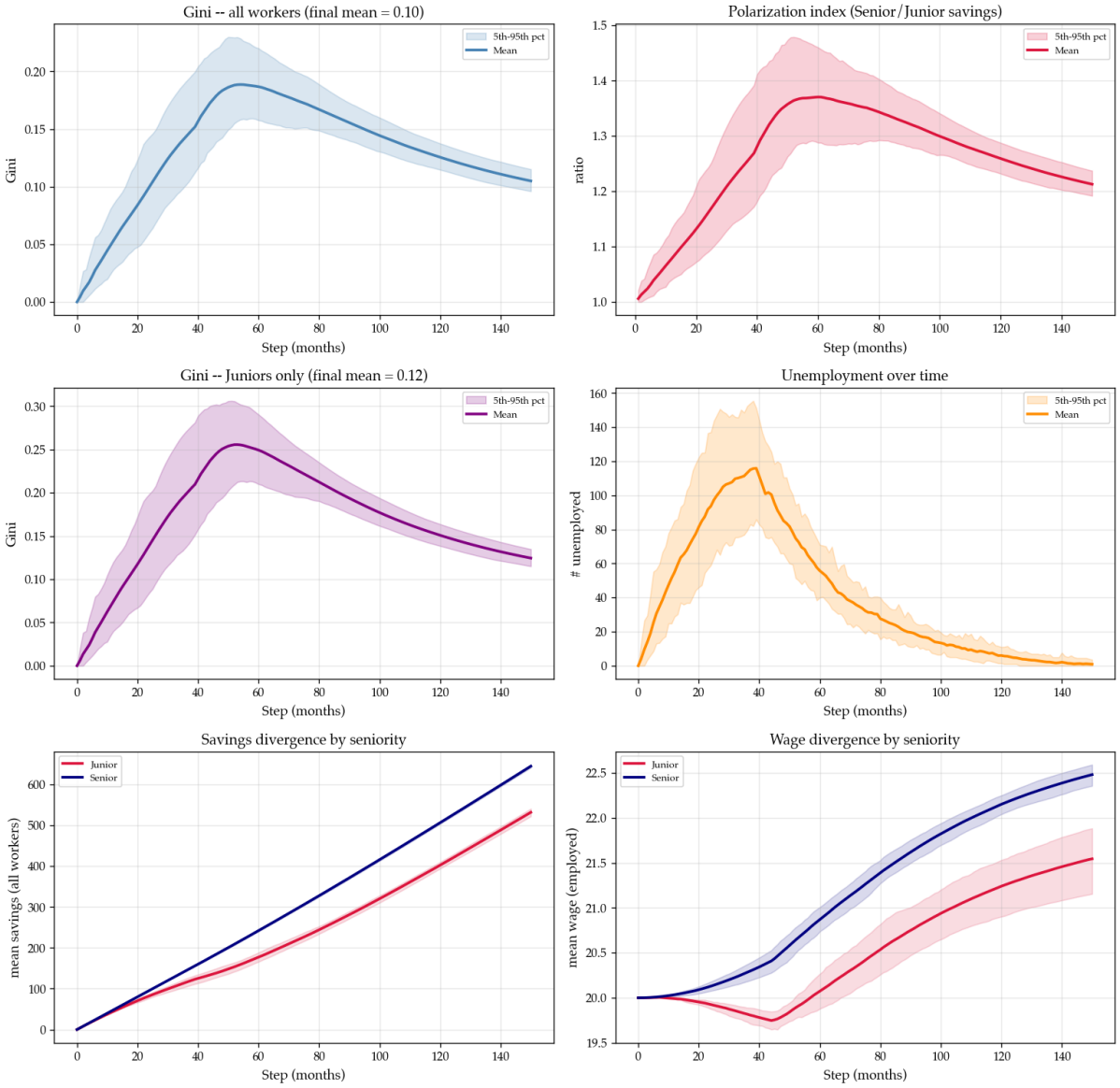


Figure 2 -- Main spec (buffer_months=3, patience_decay=0.05)

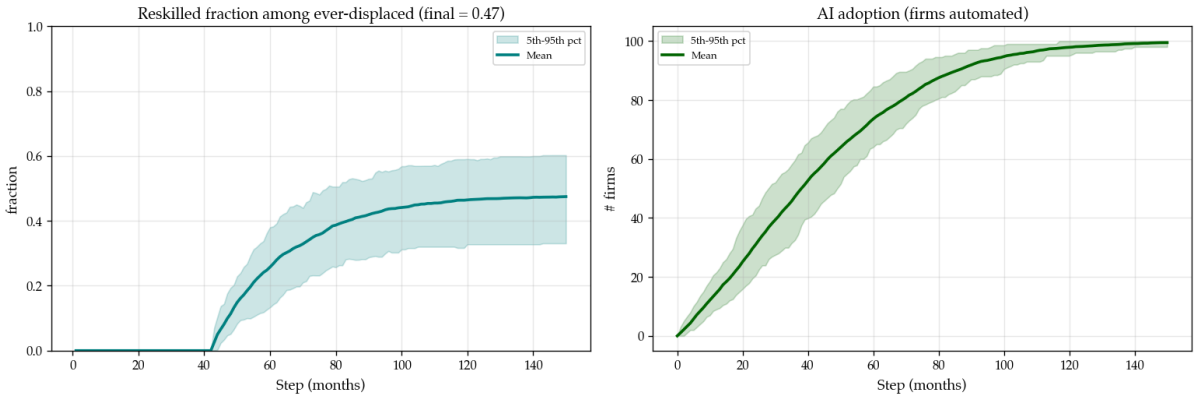


Table 2: Monte Carlo descriptive statistics. MC runs = 50, step 150

Variable	**Avg.**	**5th Percentile**	**95th Percentile**
Gini Coefficient	0.105	0.096	0.115
Polarization Index	1.212	1.192	1.237
Junior Savings Gini	0.124	0.115	0.135
Num. of Unemployed	0.90	0.00	3.55
Mean Savings (Junior)	532.14	522.42	540.97
Mean Savings (Senior)	645.10	641.58	648.78
Mean Wage (Junior)	21.55	21.16	21.88
Mean Wage (Senior)	22.48	22.36	22.59
Reskilled among displaced (%)	47.48%	33.07%	60.24%
Automated Firms (%)	99.46%	98.00%	100.00%

Note: Main model specification with parameters $B_{\text{months}} = 3$ and $\delta = 0.05$. Descriptive statistics are computed across 50 independent Monte Carlo seeds at simulation step 150.

In the main specification, as shown in Figure 1, the trajectory of the aggregate Gini coefficient follows a hump-shaped path: starting from zero (all workers begin with identical savings), it climbs to a peak of 0.189 around step 54 - the height of the automation-induced displacement wave - and then recedes to a terminal 0.105 as displaced workers are rehired and rebuild savings. Lowering the patience decay to $\delta = 0.02$ shifts the peak both later and higher (0.238 at step 62) and, crucially, raises the terminal level to 0.158: when firms stay selective for longer, the rehiring phase is slower and inequality does not fully unwind. Removing the buffer ($B_{\text{months}} = 0$) works in the opposite direction, lowering both the peak (0.170) and the terminal value (0.088), because a lower retraining threshold lets more workers reskill and re-enter employment. The two effects are roughly additive: the most unequal regime is $(\delta = 0.02, B_{\text{months}} = 3)$ and the least unequal is $(\delta = 0.05, B_{\text{months}} = 0)$.[14]

These dynamics of the Gini coefficient over the time span considered suggest that, as more firms adopt AI and thus begin displacing workers - with the probability of displacement distributed

[14]See Appendix A for the plots of the alternative model specifications mentioned.

unequally between Junior and Senior workers - inequality among workers increases. Still, as firms gradually resume hiring, prioritising reskilled workers and absorbing non-reskilled ones according to the patience-decay parameter, the Gini coefficient declines, since re-employed workers earn a wage and save accordingly, but it does not go back to its initial level. Indeed, at the beginning of the simulation all workers are equal by construction, so that the inequality we observe is an emergent feature of the economic system, stemming from the automation dynamics.

The Gini coefficient measured exclusively within the Junior cohort depicts a similar dynamics of inequality evolution. Nonetheless, inequality is overall higher when measured only among Junior workers. Indeed, since Junior workers are more likely to be fired ($p_j > p_s$), they have a different retraining path: those displaced late in the adoption wave have accumulated sufficient savings to enter retraining and are re-absorbed in the job market at full wages, while those displaced early are structurally confined in the poverty trap scenario and permanently penalised via lower wages. In stark contrast, Senior workers virtually never experience technological displacement due to their structurally low probability of dismissal, thus we expect within-group inequality for the Senior cohort to remain minimal. Restricting attention to Juniors sharpens the picture: the within-cohort Gini peaks at 0.256 (step 53) and ends at 0.124 in the baseline. This series is the most sensitive to δ: at $\delta = 0.02$ the peak reaches 0.326 and the terminal value 0.201 - a 60% increase over baseline - reflecting the divergence between Juniors who reskill quickly and those who remain displaced. Removing the buffer compresses the distribution (0.229 peak, 0.101 final), consistent with more Juniors being able to afford retraining.

Figure 1 also reports the trajectory of the Polarization Index $P(t) = \mu_t^S / \mu_t^J$. This latter, differently from the Gini coefficient, aims at evaluating the gap between junior and senior workers' average savings, as belonging to two distinct categories, rather than measuring the dispersion of individuals' savings. Starting at unity by construction, it then rises to 1.370 at step 60 before settling at 1.212; because Juniors face a much higher displacement exposure than Seniors (0.60 vs. 0.02), the gap never returns to one. The comparative statics mirror the Gini: $\delta = 0.02$ raises the peak to 1.500 and the terminal ratio to 1.309: if employers are more patient, they will be less prone to hire unskilled workers, and since junior workers are more likely to be fired, polarization remains higher. Instead, $B = 0$ lowers them to 1.326 and 1.172 respectively. Here too the adverse corner is $(\delta = 0.02, B_{\text{months}} = 3)$, which combines prolonged Junior unemployment with a high reskilling barrier.

Unemployment is sharply transient. In the baseline it spikes to roughly 116 workers around step 39 and then collapses to a near-frictional terminal level of under one worker (0.9 on average, with a 95th percentile of 3.6). The patience decay governs the *persistence* of this friction rather than its onset: at $\delta = 0.02$ the peak rises to ~ 177 workers but, far more strikingly, the terminal level jumps to 21.4 - a stock of long-term unemployed that never clears within the horizon. Removing the buffer lowers the peak (~ 94) and keeps the terminal level low (0.6), since faster reskilling returns workers to the hiring pool. The combination $(\delta = 0.02, B_{\text{months}} = 0)$ leaves an intermediate residual of 17.8. We recall that the simulation is modeled so that unemployment is not the emergent phenomenon to be studied: indeed, at the end of the simulation, all agents are rehired, although at different wage levels. The distribution was here plotted only to highlight

the dynamics of unemployment in time, raising as more firms adopt AI and then decaying as patience is eroded, rather than the level reached at the end of the simulation.

In the same figure, the mean savings trajectories of Junior and Senior workers are plotted. Both groups begin from zero and accumulate savings through wage income; the saving gap emerges as AI adoption takes off, with Junior workers being disproportionately displaced with respect to Seniors. By step 150, mean Senior savings reach approximately 645 model units against 532 for Juniors. This divergence is driven by two compounding forces. First, as mentioned, the displacement asymmetry: with $p_J = 0.60$ against $p_S = 0.02$, Junior workers face a substantially higher probability of having their savings accumulation interrupted by unemployment. Second, the underemployment penalty: Junior workers re-hired without retraining receive only $\psi = 85\%$ of the market wage, reducing their savings accumulation rate relative to both Senior workers and retrained Junior workers for the remainder of their employment spell.

Senior mean savings are almost invariant across specifications (terminal values between 644 and 652), because Seniors are largely shielded from displacement. Junior savings, by contrast, change significantly across alternative specifications: they end at 532 in the baseline but fall to 492 under $\delta = 0.02$ (selective hiring depresses Junior earnings) and rise to 556 under $B_{\text{months}} = 0$ (more Juniors reskill into higher-productivity, profit-sharing firms). The Senior-Junior savings gap therefore widens under low patience and narrows when the buffer is removed.

The wage divergence of Juniors *vis-à-vis* Seniors in the adjacent plot is notably more contained than the savings one. In the main specification mean Junior wages stabilise at approximately 21.5 model units against 22.5 for Seniors - a gap of roughly 4%. This small wage gap relative to the much larger savings gap (Seniors hold about 21% more savings than Juniors, i.e. a polarization ratio of 1.21) reflects the dynamic amplification mechanism of the model: the wage penalty affects the flow of savings accumulation, but the stock of savings compounds the disadvantage over time. A worker who receives 85% of the market wage for fifty steps accumulates far less than what they would have saved at full wages, because they also miss the additional savings that would have been earned in the interim. This non-linearity between the wage gap and the savings gap is a central distributional finding of the model. Across all four configurations the effect stays muted at the wage level: terminal Junior wages range only from 21.5 to 22.4 and Senior wages from 22.5 to 22.7, so the direct wage gap never exceeds about 4%. The buffer has a slightly larger effect on wages than δ does - $B_{\text{months}} = 0$ lifts Junior wages toward $22.2 - 22.4$ - but the key point is that the modest wage differential compounds, through the buffer-stock savings rule, into the much larger savings and polarization gaps documented above.

Figure 2 displays two additional graphs. The first one plots the fraction of ever-displaced workers who completed retraining over the period of 150 steps. This fraction rises from zero as the first retrained workers re-enter the labour market after $N_{\text{train}} = 3$ steps, eventually stabilising at approximately 0.48, implying that roughly 52% of all displaced workers are unable to afford retraining and are permanently confined in the poverty trap. Under the main specification, the retraining threshold is $\Theta = 150$ model units, which requires approximately 38 steps of uninterrupted employment to accumulate from zero savings at the baseline wage. Workers displaced before step 38 - a large share given that 47% of firms have adopted by step 36 - are therefore

structurally excluded from retraining regardless of their own preferences. This mechanical constraint is the primary driver of the poverty trap. The precautionary buffer $B_{\text{months}} = 3$ raises the threshold above its bare minimum, tightening the constraint; the sensitivity of this finding to B_{months} is examined in the Appendix.

This is the distribution most cleanly governed by the buffer, and it is where δ matters least. The reskilled fraction rises monotonically to a terminal 0.475 in the baseline, essentially unchanged at 0.481 when δ is lowered - patience decay affects who gets hired, not who can afford to retrain. The buffer, by contrast, moves it decisively: with $B = 0$ the threshold Θ falls from 150 to 90, and the reskilled fraction jumps to 0.678–0.693. This is the mechanism behind the inequality results: a lower retraining barrier is what allows displaced Juniors to re-enter employment and rebuild savings.

The second graph from Figure 2 shows the AI adoption S-curve. Adoption follows the Bass (1969) diffusion process: the imitation term $\gamma F(t)$ keeps early adoption slow, then amplifies it as the share of automated firms crosses roughly 30–40%, generating the characteristic convex take-off before saturation. By step 36 (September 2025), adoption reaches approximately 47%, broadly consistent with evidence on AI diffusion documented by Brynjolfsson et al. (2025a) used to calibrate the parameters; by step 150, adoption is essentially complete across all 100 firms. Note that the unemployment series displayed in Figure 1 mirrors the AI adoption curve with a short lag: displaced workers flood the job search pool as firms adopt and fire, producing a sharp peak in unemployment around step 39, concentrated between steps 20 and 60, followed by gradual absorption - extending to roughly step 100 - as automated firms re-enter the labour market to restore their workforces. By the end of the horizon, mean unemployment stabilises near zero, confirming that the model does not generate permanent structural unemployment but rather a prolonged transition period during which the distributional implications of the technological shock accumulate.

The trajectory of AI adoption is the sole remaining effectively invariant to changes in both δ and B_{months} within the robustness checks. In every tested configuration, the share of automated firms saturates near 99% by the final step, yielding terminal cross-seed means between 99.2% and 99.5%, with the 95th percentile reaching full saturation at 100% across all cells. This is by construction: the parameters δ and B_{months} govern the structural labour market responses to displacement, leaving the Poisson process with competitive imitation that drives technology diffusion entirely untouched. The AI diffusion process is therefore common to all considered scenarios, and the divergences documented above with respects to other variables are entirely distributional consequences of how the labour market absorbs a near-identical automation shock.

Lower patience decay ($\delta = 0.02$) prolongs unemployment and depresses Junior earnings, raising every inequality measure and leaving a persistent stock of long-term unemployed; a higher buffer ($B_{\text{months}} = 3$) raises the retraining threshold and suppresses re-skilling, which is the deeper driver of the terminal savings gap. They compound rather than offset: $(\delta = 0.02, B_{\text{months}} = 3)$ is the most adverse regime on inequality, polarization, and residual unemployment, while $(\delta = 0.05, B_{\text{months}} = 0)$ is the most benign and delivers the highest re-skilling - all against a common, near-complete automation path.

5. Conclusions

This paper developed an ABM modelling AI adoption in a bipartite firm-worker economy in the spirit of the Boltzmann wealth model. We retain the features that make the Boltzmann framework a laboratory for distributional analysis - an initially homogeneous population of agents and a closed monetary circuit in which wages and consumption expenditure recirculate without leakage - but we deliberately switch off its inequality-generating engine, the stochastic bilateral exchange that redistributes wealth at random between agents. In our baseline, equal endowments and uniform wages instead constitute a stable fixed point, so that every distributional movement we observe can be attributed exclusively to the two mechanisms we add: the seniority-asymmetric AI-adoption shock and the Axelrod imitation dynamics that propagate it.

In this way, every distributional movement we observe can be attributed exclusively to the technological shock and the labour-market mechanisms it triggers. Adoption diffuses through an Axelrod mean-field imitation channel that reproduces the Bass (1969) S-curve, and its labour-market consequences are governed by a seniority-asymmetric displacement rule, a profit-sharing wage schedule, and a savings-constrained retraining pathway calibrated to the early-career employment losses documented by Brynjolfsson et al. (2025a).

Our first research question asked how staggered firm-level adoption drives endogenous polarization and inequality between Junior and Senior workers. The model delivers a clear answer: inequality is a transient but nonself-correcting phenomenon. Starting from a perfectly equal population, the Gini coefficient, the Junior-only Gini coefficient and the Polarization Index all rise sharply during the diffusion wave, peak near the middle of the horizon, and then only partially recede as displaced workers are reabsorbed. At period 150, all the variables display a terminal level well above their neutral starting point. The mechanism responsible for that is asymmetric exposure: with a Junior displacement probability an order of magnitude larger than the Senior one, the adoption shock falls almost entirely on early-career workers, whose interrupted savings trajectories translate a modest terminal wage gap into a substantially larger gap in accumulated savings. This flow-to-stock amplification, operating through the buffer-stock savings rule, is the model's central distributional finding: small, persistent differences in earnings compound into sizeable and durable disparities in wealth.

Our second research question concerned the interaction of labour market frictions, credit constraints and displacement timing in determining unemployment, access to retraining and entry into a path-dependent poverty trap. Here the two frictions we vary in sensitivity analysis act through distinct channels. The hiring-selectivity parameter δ governs the persistence of unemployment: lower patience decay keeps firms selective for longer, prolonging Junior joblessness and leaving a residual stock of long-term unemployed that does not clear within the horizon, while leaving the reskilling rate essentially untouched. The precautionary buffer B_{months}, by contrast, governs access to retraining: by raising the savings threshold Θ, it mechanically excludes workers displaced early, before they have accumulated enough to enroll, from the retraining pathway, regardless of their preferences. Because early displacement coincides with the steepest part of the adoption curve, this timing constraint, rather than worker choice, is the primary engine of the poverty trap: workers caught early are pushed into re-employment at

an underemployment penalty that slows their saving and makes subsequent escape progressively harder. The two frictions compound rather than offset, so that the regime combining low patience with a high buffer is the most adverse on every distributional margin, while its opposite is the most benign and delivers the highest reskilling - all against a near-identical, near-complete automation path.

Taken together, these results carry a policy-relevant implication that follows directly from the model's structure. Because diffusion is effectively invariant to the labour-market frictions - since automation saturates in every scenario - the distributional outcome is determined not by whether firms adopt but by *how* the labour market absorbs the shock. The binding constraint on recovery is liquidity at the moment of displacement: the workers who fall into the poverty trap are not those unwilling to retrain but those without the savings to afford it when the shock arrives. This points to the relaxation of the credit constraint on retraining - rather than the slowing of adoption - as the margin on which intervention would be most effective, a reading consistent with the model's finding that lowering the retraining threshold sharply raises re-skilling and compresses terminal inequality.

These conclusions should be read in light of the model's deliberate simplifications. The supply side is intentionally stylised: firms are ex-ante identical, adopt through a common Poisson-plus-imitation hazard, and face no heterogeneity in size, sector, or technology, so the model speaks to the redistributive consequences of adoption rather than to firms' investment decisions. This simplification can still be admissible in light of a methodological choice: that of relying on an empirical study, Brynjolfsson et al. (2025a), that takes into account only two sectors, deemed as more harshly shaked by AI technology adoption, namely those of software and services to the client.

We model only the recurring per-step cost of AI and fold the one-off setup cost into initial conditions, we abstract from a topological adoption network by taking the complete-graph (mean-field) limit, and we treat the Junior *vis-à-vis* Senior distinction as binary, a coarse proxy for a continuous tenure distribution. Several of these are natural avenues for future work: introducing firm heterogeneity and an explicit adoption network, adding a transition cost at the moment of adoption, endogenising the profit-sharing and displacement parameters, and enriching the retraining technology beyond a single binary threshold.

Extending the framework in these directions would test the robustness of the poverty-trap mechanism identified here and sharpen its quantitative implications. The qualitative lesson, however, is likely to survive them: because automation diffuses to near-completion in every scenario we examine, the distributional cost of AI is determined not by *whether* firms adopt, but by *how* the labour market absorbs the shock, and, in particular, by whether displaced workers hold enough liquidity to afford retraining at the moment they are hit. It is the financial fragility of the workers the shock displaces, rather than any feature of the adoption process itself, that governs how a transient disruption translates into long-lasting inequality.

A. Alternative Model Specifications

Figure 1 -- buffer_months=3, patience_decay=0.02

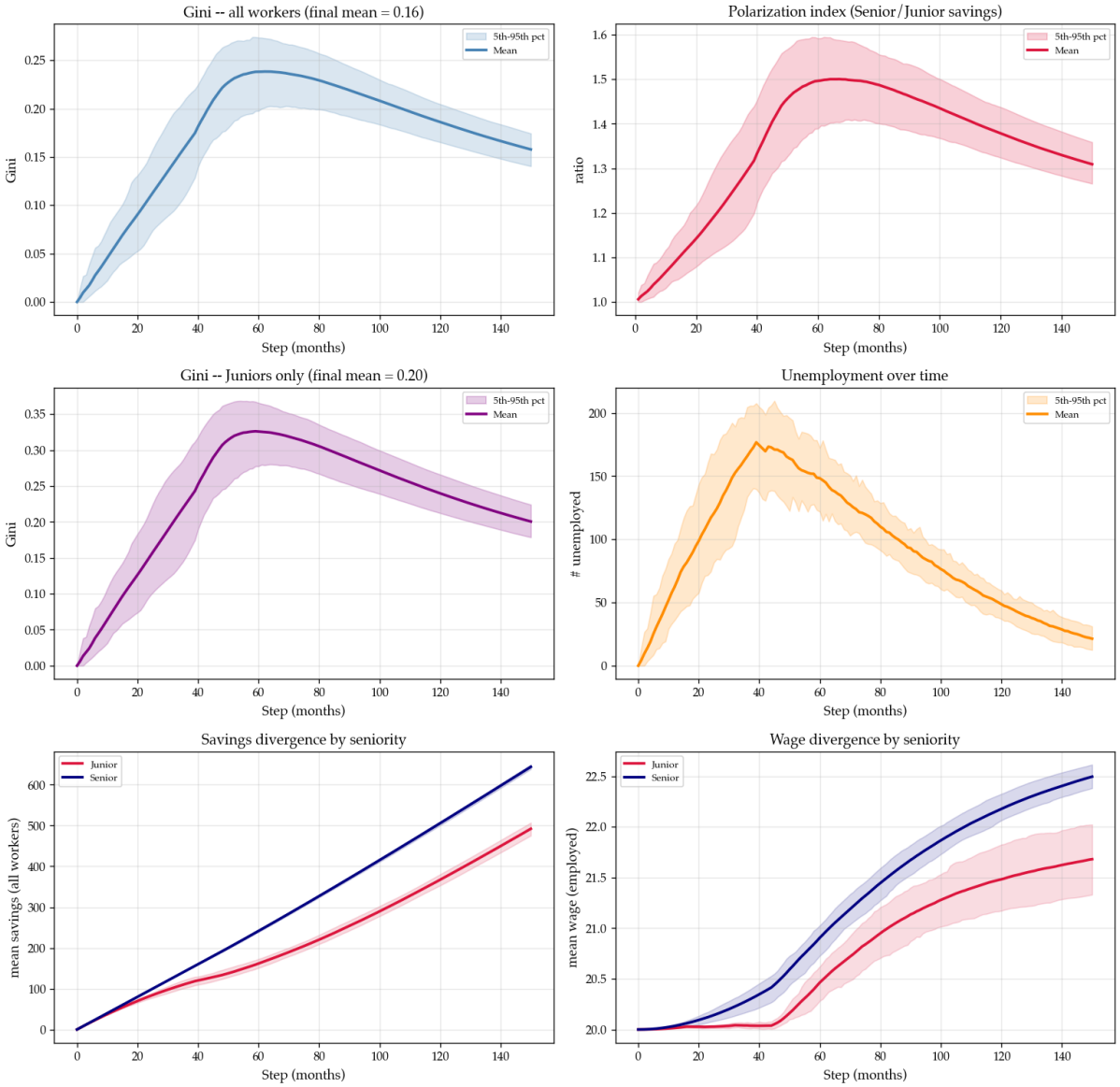


Figure 2 -- buffer_months=3, patience_decay=0.02

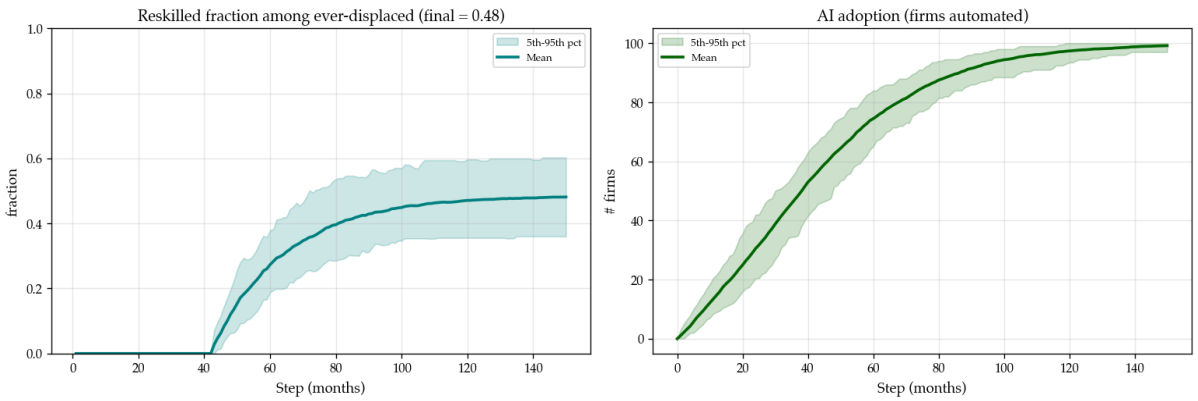


Figure 1 -- buffer_months=0, patience_decay=0.05

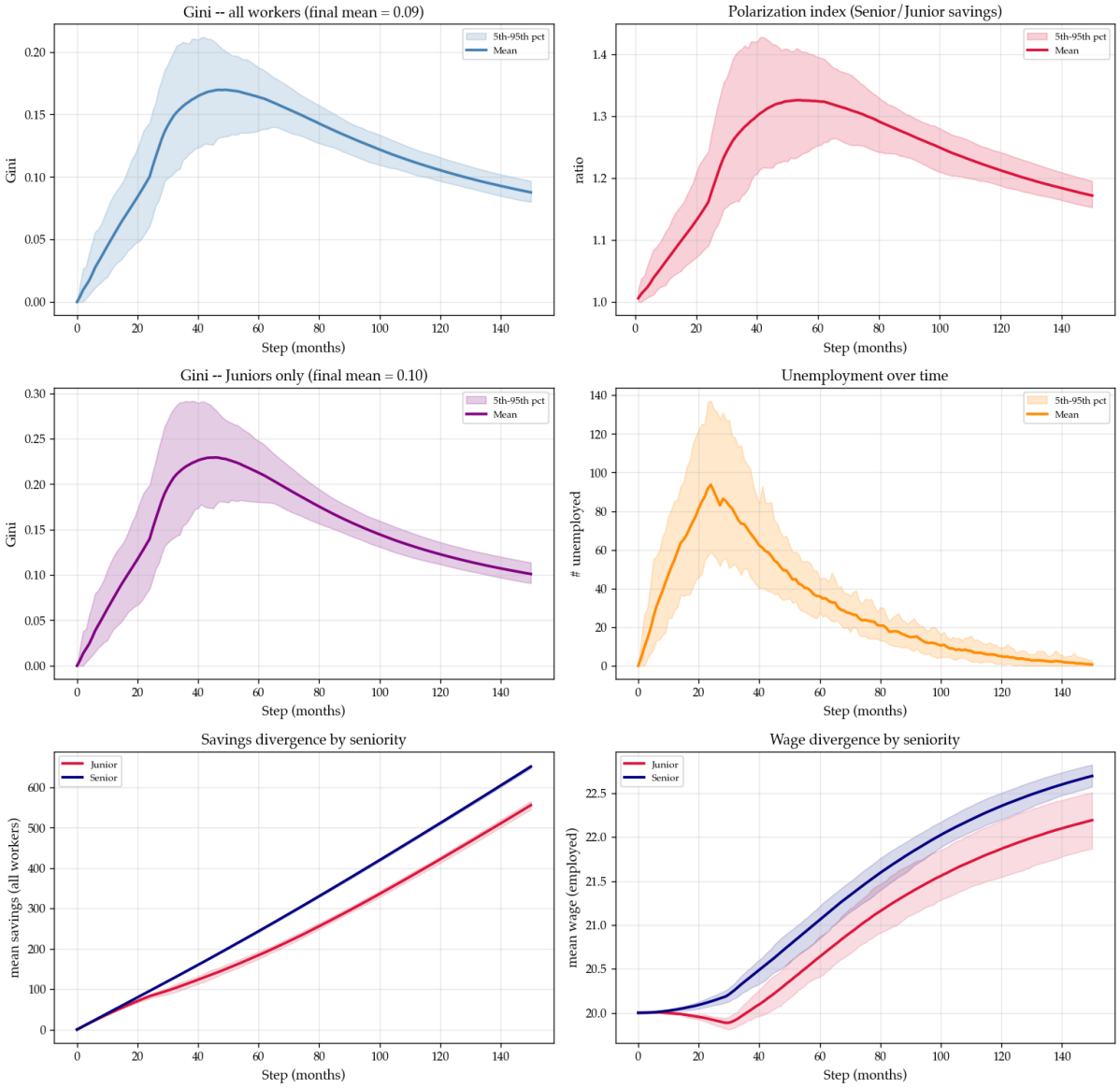


Figure 2 -- buffer_months=0, patience_decay=0.05

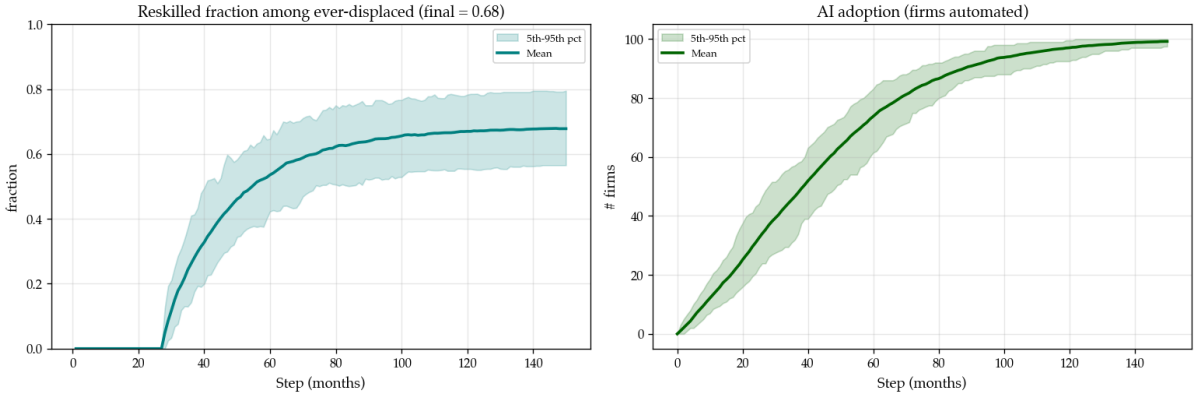


Figure 1 -- buffer_months=0, patience_decay=0.02

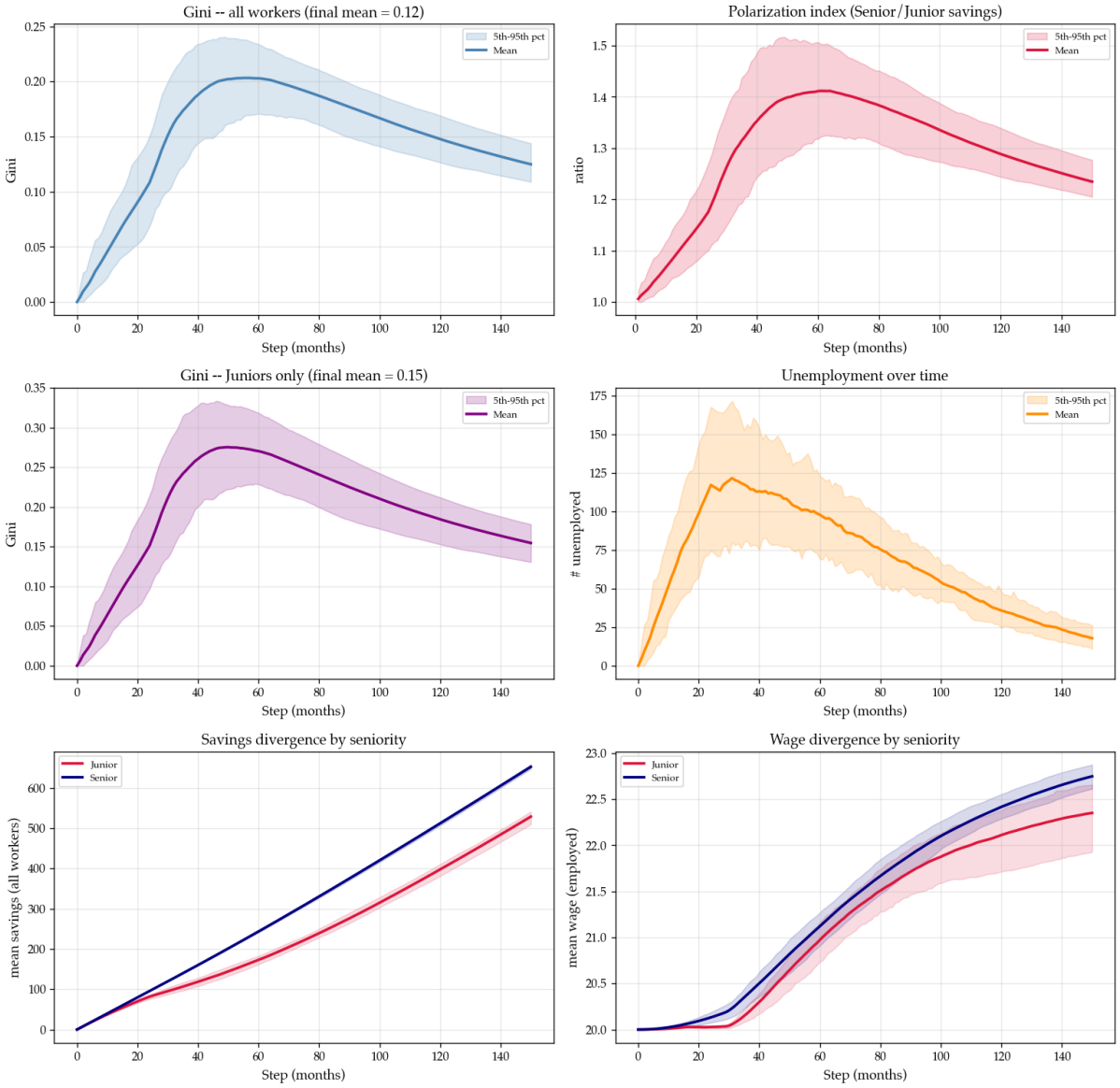
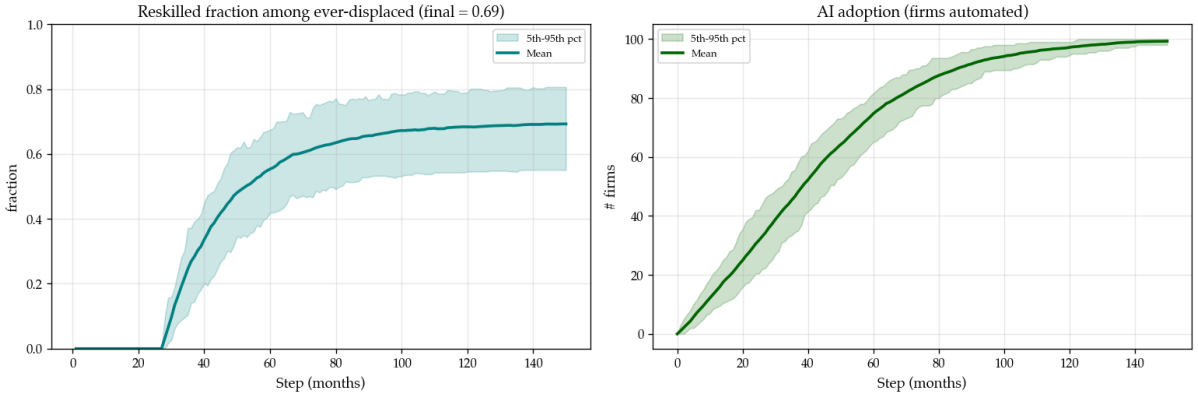


Figure 2 -- buffer_months=0, patience_decay=0.02



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